AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Mathematics (Science Group) Paper II

Time: 1 hour 40 minutes Marks: 30

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

Model U-EB for Teacher Paper 2027 Learning only

INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.
3. There are NINE questions in this exam. Answer ALL questions. Questions 4, 5 and 7 each offer TWO choices. For each of these questions, attempt ONE choice only. Question 9 offers THREE choices. For this question, attempt ANY TWO only.
4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.
5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.
6. DO NOT mention your name or school name anywhere inside the paper.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a simple calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meaning.
- The same formulae will be provided in the annual and september examinations.

Real Numbers

$x^m \times x^n = x^{m+n}$ $\quad (x \times y)^n = x^n \times y^n$ $\quad \left(x^m\right)^n = x^{mn}$

$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$, where $y \neq 0$ $\quad \frac{x^m}{x^n} = x^{m-n}$

Logarithms

$\log_a(m \times n) = \log_a m + \log_a n$ $\quad \log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$ $\quad \log_a b = n \Leftrightarrow a^n = b$

$\log_a(m)^n = n\log_a m$ $\quad log_m n = \frac{log_a n}{log_a m}$

Sets Theory

$A \Delta B = (A \cup B) - (A \cap B)$ $\quad (A \cup B)^c = A^c \cap B^c$ $\quad (A \cap B)^c = A^c \cup B^c$

Factorisation

$(a+b)^2 = a^2 + 2ab + b^2$ $\quad (a-b)^2 = a^2 - 2ab + b^2$

$(a+b)^2 + (a-b)^2 = 2\left(a^2 + b^2\right)$ $\quad (a+b)^2 - (a-b)^2 = 4ab$

$(a+b)(a-b) = a^2 - b^2$ $\quad (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ $\quad (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$a^3 + b^3 = (a+b)\left(a^2 - ab + b^2\right)$ $\quad a^3 - b^3 = (a-b)\left(a^2 + ab + b^2\right)$

Introduction to Coordinate Geometry

$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $\quad \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$y = mx + c$ $\quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Similar Figures

$\frac{1}{2}h \times b$ $\quad l \times b$ $\quad l^2$ $\quad 6l^2$

l^3 $\quad \pi r^2$ $\quad l \times b \times h$ $\quad 2(l \times b + b \times h + l \times h)$

Introduction to Trigonometry

$1° = \frac{\pi}{180} rad$ $\quad 1rad = \left(\frac{180}{\pi}\right)^\circ$ $\quad H^2 = P^2 + B^2$

$\sin^2\theta + \cos^2\theta = 1$ $\quad 1 + \tan^2\theta = \sec^2\theta$ $\quad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$

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Q.1. (Total 3 Marks)

Apply the laws of exponents to simplify $\sqrt[3]{125 \times \frac{x^{-4}}{x^2}} \times \sqrt{\frac{x^4}{25}}$ in its simplest form.

Q.2. (Total 3 Marks)

Find the value of x for $\log_6 1 + \frac{\log_6 216}{\log_6 36} = x$.

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Q.3. (Total 3 Marks)

In a class of 50 students, 25 students use *X*-mobile network; whereas, 38 students use *Y*-mobile network. Hence, each student uses at least one of the two mobile networks.

Calculate the number of students who use both *X* and *Y*-mobile networks.

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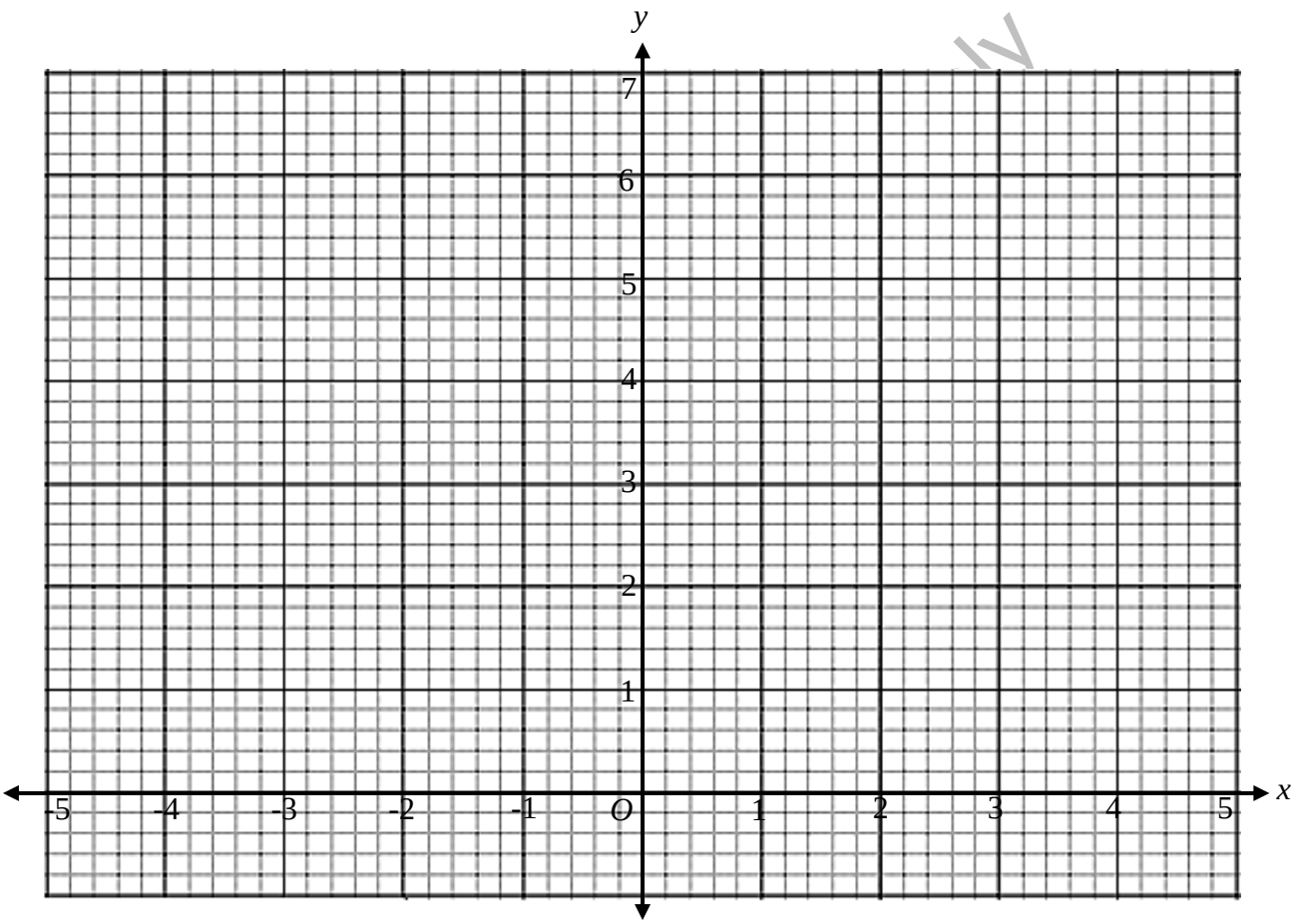
(ATTEMPT EITHER PART a OR PART b OF Q.4.)

Q.4. (Total 3 Marks)

a.

i. By completing the given table, plot the function $f(x) = |x - 2| + 3$, on the given graph. (2 Marks)

x	0	2
$f(x)$	5	



ii. Write the range of the function given in part i. (1 Mark)

_____

_____

(ATTEMPT EITHER PART a OR PART b OF Q.4.)

b. Two sets are defined as $S = \{1, 2, 3, 4\}$ and $T = \{2, 6, 8\}$.

i. Find an into function from S to T. (1 Mark)

ii. Is it possible to find a one-one and onto function (bijective function) from S to T?

Justify your answer. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 3 Marks)

a. The algebraic expression $x^3 + 3x^2 + ax - b$ is divided by $x + 2$ and the remainder is -12.

Find the value of a when b = – 12.

b. Factorise the expression $36a^4 + 12a^2b + b^2 - 9$.

Q.6. (Total 3 Marks)

Complete the solution of the square root of an algebraic expression by filling the space provided.

.....	$36x^4 - .....x^3 + ...x^2 - 16x + ...$
.....	.....
$12x^2 - .....x$	$- 96x^3 + ...x^2$
$- .....x$	$\mp 96x^3 \pm .....x^2$
$12x^2 - 16x + 1$	$\times \quad 12x^2 - .....x + ...$
	$.....x^2 - 16x + ...$

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 3 Marks)

a. Given that $k - k^2\sqrt{x+1} = k\sqrt{x+1}$, where $k > 0$.

Show, by working, that the above equation reduces to $x+1 = \left(\frac{1}{k+1}\right)^2$

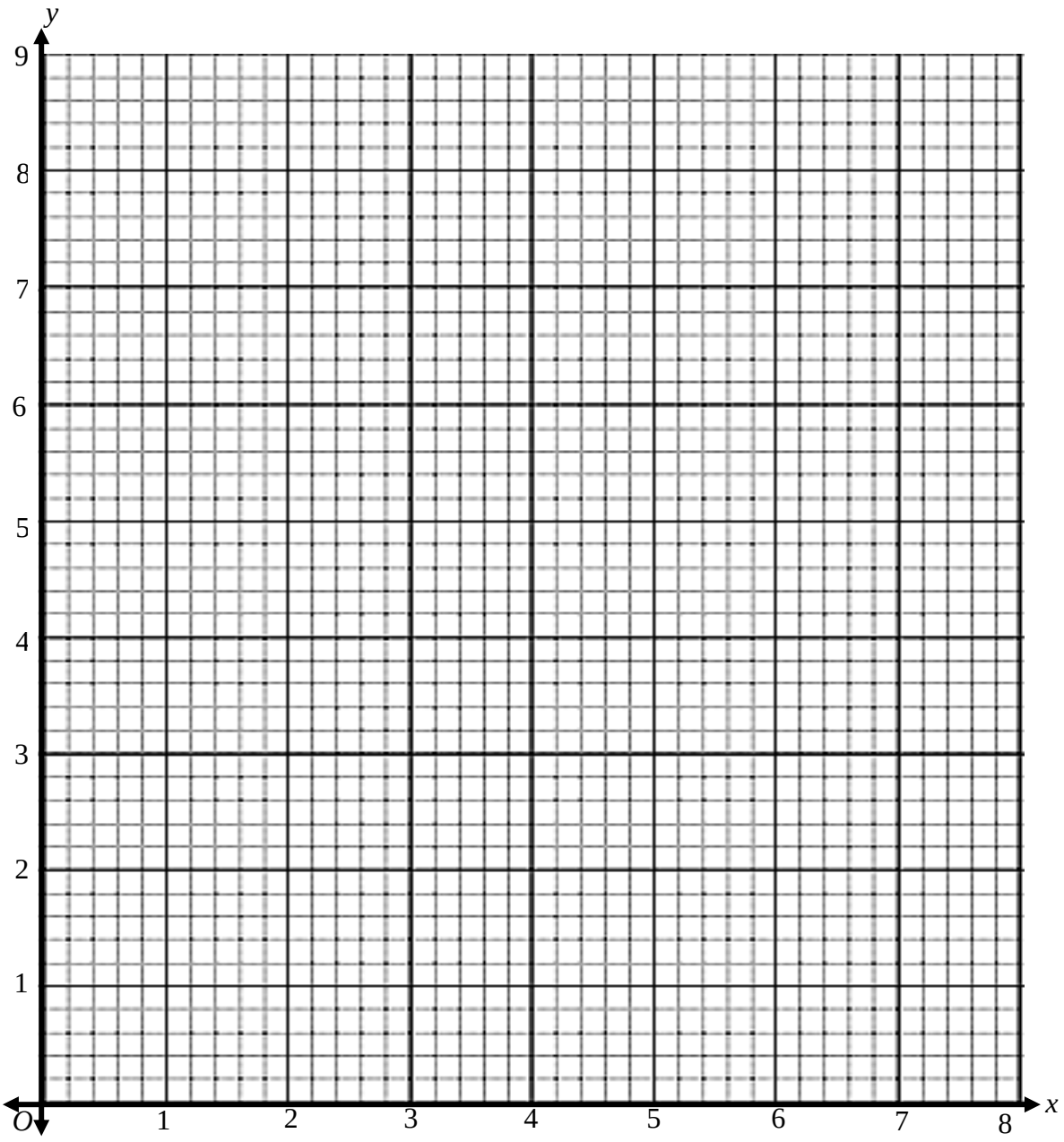
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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

b. Draw the graph for the inequalities $2x + y < 8$ and $x + y < 6$.

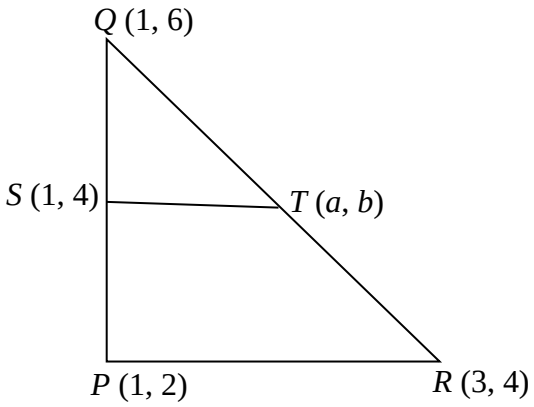
Hence, find the region bounded by the inequalities.

(**Note**: $x > 0$ and $y > 0$)



Q.8. (Total 3 Marks)

The coordinates of the vertices of a triangle PQR are P (1, 2), Q (1, 6) and R (3, 4). The midpoints of PQ and QR are S (1, 4) and T (a, b) respectively.



NOT TO SCALE

Find the coordinates of point T and distance between S and T.

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.9.)

Q.9. (Total 6 Marks)

a. A man stands on a flat surface at a distance of 50 metres from a vertical x metre high electric pole. The angle of elevation of the top of electric pole from the man on the flat surface is θ.

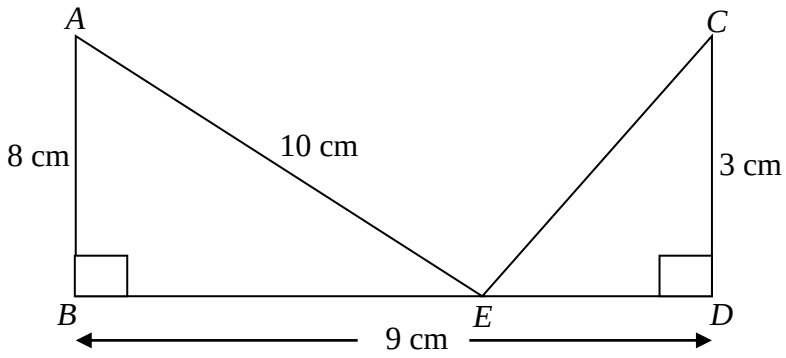
i. Show that $\theta = \tan^{-1}\left(\frac{x}{50}\right)$. (1 Mark)

ii. Hence, find the height of electric pole when the angle of elevation is 45°. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.9.)

b. In the given diagram, *ABE* and *CDE* are two right angled triangles.



NOT TO SCALE

Calculate the length of *CE*. (3 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.9.)

c. Prove that $\frac{\sin\theta}{\tan\theta+\cot\theta}=\sin^2\theta\cos\theta$ (3 Marks)

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END OF PAPER

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